Bees 3: a bumblebee population

Simon Frost

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```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Plots
```

Bumblebees live differently from honey bees. A colony lasts one season: a queen who hibernated alone founds it in spring, it rears workers, then males and new queens, and then dies; only the new queens overwinter. What persists from year to year is the *population* of colonies, so that is what this model follows, in numbers per square kilometre.

Three modules

The life cycle is assembled from three nets that share a list of stages: what queens do, what the colony rears, and how long adults live.

```
p = BUMBLEBEE_PARAMETERS
stay = 1 - p.queen_mortality
stages = [:hibernating, :searching, :founding, :growing, :reproducing,
 ↪  :ending,
          :worker_brood, :worker, :male_brood, :male, :gyne_brood,
           ↪  :new_queen]

queens = ValuedProjectionNet(stages,
    :dormancy => [(:hibernating => :hibernating) => 1.0],
```

```
    :waiting => [(:new_queen => :new_queen) => 1.0],
    :overwintering => [(:new_queen => :hibernating) => 1.0],
    :emergence => [(:hibernating => :searching) => p.winter_survival],
    :searching => [(:searching => :searching) => 1 - p.nest_finding -
     ↪  p.search_mortality],
    :nesting => [(:searching => :founding) => p.nest_finding],
    :first_workers => [(:founding => :growing) => 1.0],
    :colony_cycle => [(:growing => :growing) => stay * (1 - p.switch),
     ↪  (:growing => :reproducing) => stay * p.switch,
                      (:reproducing => :reproducing) => stay * (1 -
 ↪  p.ending), (:reproducing => :ending) => stay * p.ending,
                      (:ending => :ending) => stay])

rearing = ValuedProjectionNet(stages,
    :first_brood => [(:founding => :worker) => p.first_workers],
    :worker_eggs => [(:growing => :worker_brood) => p.eggs],
    :sexual_eggs => [(:reproducing => :gyne_brood) => p.gyne_fraction *
     ↪  p.eggs,
                     (:reproducing => :male_brood) => (1 - p.gyne_fraction)
 ↪  * p.eggs],
    :eclosion => [(:worker_brood => :worker) => p.brood_survival,
     ↪  (:male_brood => :male) => p.brood_survival,
                  (:gyne_brood => :new_queen) => p.brood_survival *
 ↪  p.gyne_survival])

adults = ValuedProjectionNet(stages,
    :worker_life => [(:worker => :worker) => 1.0], :male_life => [(:male =>
     ↪  :male) => 1 - p.male_mortality])

life = queens ⊕ rearing ⊕ adults
transition_names(life)
```

```
14-element Vector{Symbol}:
 :dormancy
 :emergence
 :colony_cycle
 :male_life
 :first_brood
 :first_workers
 :waiting
 :worker_life
 :nesting
 :eclosion
 :searching
 :sexual_eggs
 :worker_eggs
```

`:overwintering`

Founding a nest takes four weeks, and worker, male and gyne brood take 22, 26 and 30 days to develop:

```
daily = expand_stages(ResponsiveProjectionNet(life), :founding => 28,
 ↪  :worker_brood => 22, :male_brood => 26, :gyne_brood => 30)
length(stage_names(daily))
```

```
114
```

`bumblebee()` is this net with its responses:

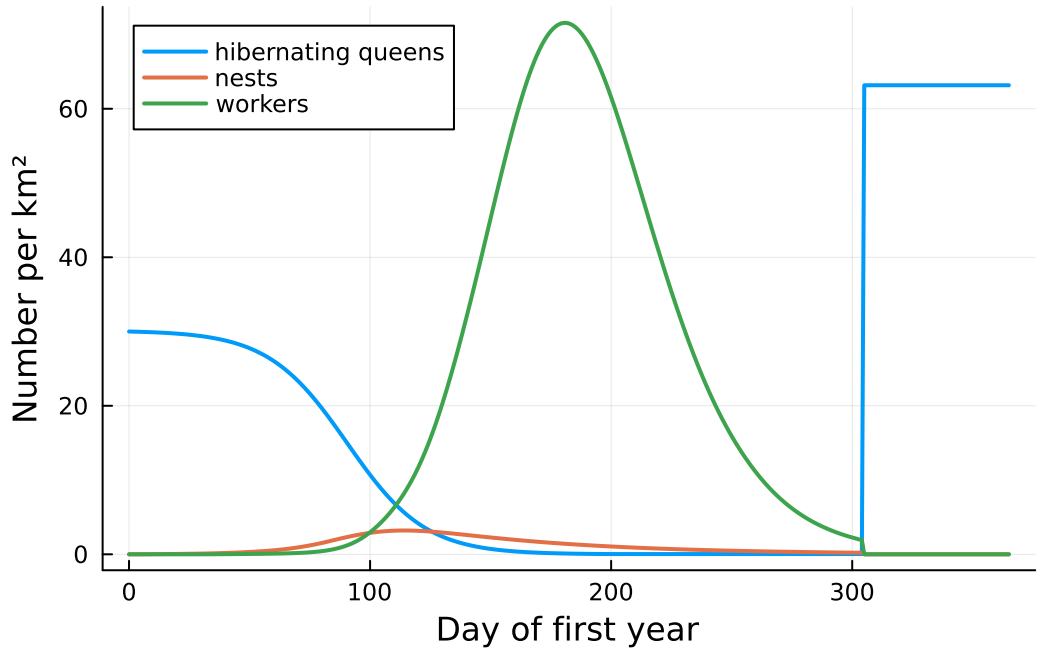
Transition	Response	Source
`emergence`, `dormancy`	queens emerge around day 91 (SD 28)	Bumble-BEEHAVE
`nesting`	only where a nest site is free	Knight et al. 2005 (about 30 nests per km²)
`worker_eggs`, `sexual_eggs`	eggs per colony rise with workers per colony, to the power 0.54	Cresswell 2017
brood ageing	larvae are lost when pollen or nectar is short; gyne larvae first	Bumble-BEEHAVE (Liebig's minimum)
`worker_life`, founding	foragers die faster when nectar is hard to find	proposed
`waiting`, `overwintering`	new queens stay dormant until day 305, then join the hibernating queens	Bumble-BEEHAVE
everything else	all bees not dormant die on day 305	Bumble-BEEHAVE

Colonies are pooled: a response sees the total number of workers and of colonies, so "workers per colony" is an average over colonies of different ages.

A season

```
bb = bumblebee()
sol = project(bb, honeybee_state(bb; queens = 30), 1:(365 * 3))
day = sol.t[1:366]
nests = stage_total(sol, :bumblebee, [:founding, :growing, :reproducing,
 ↪  :ending])
```

```
plot(day, [stage_total(sol, :bumblebee, :hibernating)[1:366] nests[1:366]
 ↪  stage_total(sol, :bumblebee, :worker)[1:366]];
     label = ["hibernating queens" "nests" "workers"], xlabel = "Day of
 ↪  first year", ylabel = "Number per km²", lw = 2)
```




```
first_year = 1:365
total(stage) = sum(stage_total(sol, :bumblebee, stage)[first_year])
founded = total(:founding28)
(established = round(founded / 30; digits = 3),
 workers_per_colony = round(total(:worker_brood22) / founded +
 ↪  p.first_workers; digits = 1),
 gynes_per_colony = round(total(:gyne_brood30) / founded; digits = 1),
 males_per_colony = round(total(:male_brood26) / founded; digits = 1))
```

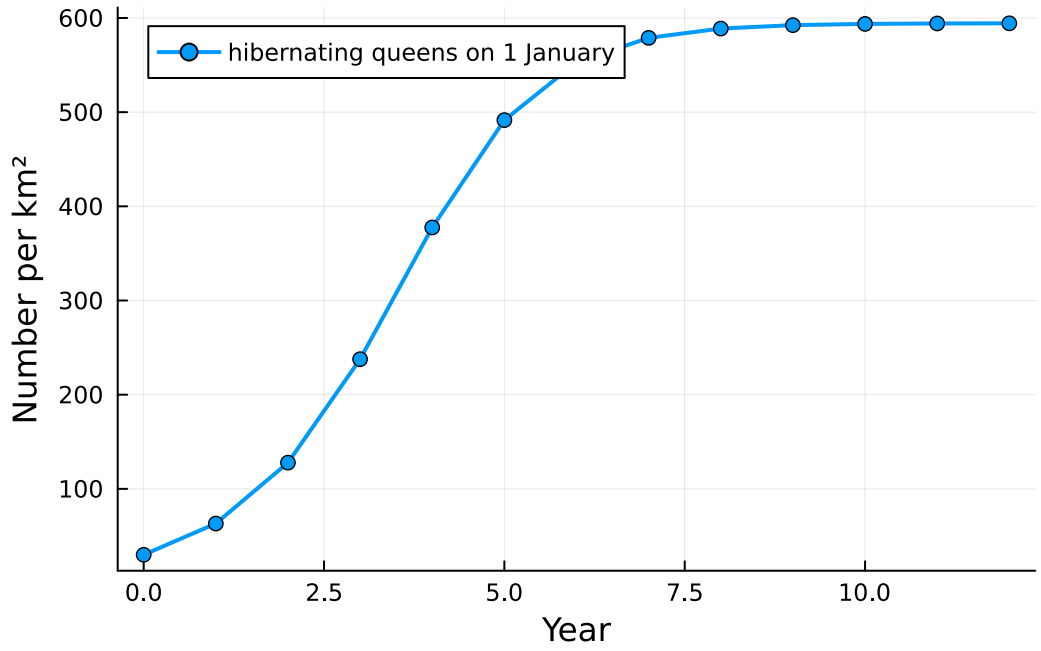
```
(established = 0.105, workers_per_colony = 82.1, gynes_per_colony = 22.3,
 ↪  males_per_colony = 27.8)
```

Published values for *Bombus terrestris*: about 0.12 of hibernating queens establish a colony; a colony rears 76 to 86 workers, 19 to 25 new queens and 22 to 30 males (Becher et al. 2018, simulated and empirical). The egg-laying rate, the rate of switching to reproduction and the duration of the reproductive phase were set to land in those ranges.

Nest sites

With forage unlimited, the population grows until nest sites run out.

```
long = project(bb, honeybee_state(bb; queens = 30), 1:(365 * 12))
start_of_year = 365 .* (0:12) .+ 1
plot(0:12, stage_total(long, :bumblebee, :hibernating)[start_of_year];
 ↪  marker = :circle, lw = 2,
     label = "hibernating queens on 1 January", xlabel = "Year", ylabel =
 ↪  "Number per km²")
```




```
maximum(stage_total(long, :bumblebee, [:founding, :growing, :reproducing,
 ↪  :ending]))
```

```
25.476473653391672
```

Adding the virus: one more module

Bumblebees have no *Varroa*. They can pick DWV up at flowers that infected honey bees have visited, and the evidence on what it does to them is mixed (the next vignette). In the model, infection is a fourth module: one more stage and two transitions, merged in with `⊕`.

```
sick = bumblebee(virus = true)
setdiff(stage_names(sick, :bumblebee), stage_names(bb, :bumblebee)),
setdiff(transition_names(system(sick)[:bumblebee]),
 ↪  transition_names(system(bb)[:bumblebee]))
```

```
([:infected_worker], [:infected_life, :infection])
```

Without honey bees there is no source of virus, so this population behaves exactly like the one above. The next vignette puts the two species on the same flowers.

What is provisional

Larval starvation rates, the extra sensitivity of gyne larvae, and the effect of nectar shortage on forager mortality are assumptions. Pooling colonies loses the large differences between colonies seen in the field (in Crone & Williams 2016 only about a quarter of colonies produced any queens). `BUMBLEBEE_PARAMETERS` and the comments in `src/bees.jl` list every value with its source.

References

Becher, M. A. et al. (2018). Bumble-BEEHAVE. *Journal of Applied Ecology* 55, 2790-2801. Cresswell, J. E. (2017). *Ecological Entomology* 42, 221-229. Crone, E. E. & Williams, N. M. (2016). *Ecology Letters* 19, 460-468. Banks, H. T. et al. (2017). *Ecological Modelling* 351, 14-23. Knight, M. E. et al. (2005). *Molecular Ecology* 14, 1811-1820.